

Introduction to Design Thinking



Outline

- Why design thinking important?
- What problems are design problems?
- Applied design thinking is strategic innovation
- A brief history and frameworks
- Double Diamond Design Methodology
- Design thinking DevOps and adoptions

Why design thinking important?

What Is Design Thinking and Why Teach It?

- Recently, a small wooden box was received as a gift. This box is beautiful and has an intricate wooden pattern. It was beautiful but seemed ornamental because it did not open.
- We thought it was a showpiece we could use as a paperweight. None of us who saw the box realized that it could be opened and that it was actually a puzzle—a problem to be solved.
- When we were finally told, it was quite difficult to figure out, and luckily, we were given a written solution key.
- Once we knew that the box could be opened and it had a solution, it was an even better gift than originally thought. It could secretly hold special items. It has that wonderful feature of being an unobvious puzzle with a solution that will work to open it every time (Figure 1-1)

Figure 1-1: Wooden Puzzle Box



What Is Design Thinking and Why Teach It?

- Finding the solutions to puzzles is a fun kind of problem-solving. Design thinking is another kind of problem-solving.
- Design thinking aims to solve problems that do not have working solutions or may be conducive to multiple possible solutions. These problems are different from puzzle problems.
- They not only can have multiple solutions that can solve the problem, they also can potentially be solved in many different ways. They can be problems such as how to design a way to secretly hide precious items, and they can be problems that are huge and complex such as those aimed at mitigating climate change.
- In this way, design thinking is thought to be used to solve “sticky” and “wicked problems”.

What Is Design Thinking and Why Teach It?

- With design thinking you are also looking to solve problems in innovative ways—by focusing intensely on possible design solutions and by coming up with solutions that might not ever have been tried before or are unique.
- The wooden gift box was obviously designed as a secret box. We thought: how clever that someone made something so beautiful to help people that needed secret places to store their precious things!
- Designing the puzzle box was itself a design thinking solution because it meets the special needs of people in a fun and creative way.

What Is Design Thinking and Why Teach It?

- Let's take a quick look at why design thinking problems are different from solving fixed-solution problems such as puzzles.
- That will help us see what is unique about design thinking and why we think it has a place in K-12 classrooms.
- We'll review the kinds of problems that are taken up with design thinking and how it is similar and different from other ways we approach and solve problems in the world and in the classroom.
- We will also introduce the processes and stages of design thinking. We will discuss the skills and mindsets we bring as designers to the problem-solving space.

What Problems Are Design Problems?

What Problems Are Design Problems?

- Design thinkers might try to solve problems for which there are no current solutions.
- The aim is to solve problems that have not already been solved or to solve existing problems in new and creative ways.
- We try to design new solutions that have not been tried before. Or sometimes we try to improve on current solutions.
- The kinds of problems we try to solve might be local ones, such as how to improve traffic flow in front of the school, or they might be more global challenges such as how to create access to clean water in deserts or in areas suffering from drought.

What Problems Are Design Problems?

- You might think you or your students are too young to approach such complex problems in the world, but we promise that you and they can be pleasantly surprised.
- We have seen kindergarteners design solutions for characters in the fairytale Cinderella, we have middle school students design an experiential museum for younger students of ancient Egyptian culture, and we have seen high school students work in partnership with a local biotech company to design patient comfort solutions.
- We have seen design projects focus on social studies, science, math, literature, social-emotional, and community topics. There are a range of problems that can be engaged in every classroom and at every level.

What Problems Are Design Problems?

- Besides coming up with solutions to problems, the actual process of engaging in design thinking helps to develop knowledge, skills, and mindsets that are important in K-12 education.
- Design thinkers are people who can solve complex problems. Through design thinking projects, students become introduced to and start developing and mastering a toolkit of skills, activities, and mindsets that they bring to problem-solving.
- They can identify problems and needs, they learn to research the problem spaces to understand current and past conditions and the ways they've been approached.

What Problems Are Design Problems?

- They engage in empathy processes to make sure that during problem-solving, they stay focused on the needs of the people (or other beings) who will be the recipients and ultimate end-users of the design.
- They learn how to create prototypes for designs, how to communicate about them with others, and how to iterate and improve them based on feedback.
- These skills and ways to apply them are important to students.

What Problems Are Design Problems?

- Besides academic skills, young design thinkers also develop important working mindsets and dispositions through design thinking. They learn how to work in teams, how to think of failure as contributing valuable lessons as part of the learning process, and how to communicate with audiences outside of their classrooms.
- There are social, emotional, and communication-related aspects to design thinking. While many 21st-century mindsets and dispositions are important to learn and might already be part of teaching and learning in your classroom, design thinking relies on an empathy orientation.
- Empathy makes design thinking distinctive and it is the foundation that makes the whole problem-solving process tick.

What Problems Are Design Problems?

- We will have much to say about empathy development throughout the book as well as other mindsets that are engaged and exercised through design thinking such as being collaborative, expecting failure to be a valid part of the learning process, having a tendency to take action, try things out, and learn to receive and pour feedback into better solutions.
- The mindset development clearly distinguishes design thinking from other kinds of problem-solving processes. It is one of the reasons why it is especially important for K-12 students to learn design thinking.

What Problems Are Design Problems?

- We think design thinking is important training for students who will pass through their schooling years into a world that is constantly changing, not totally predictable, and will require comprehensive problem-solving at home, at work, and in the community. It helps students build their problemsolving toolkits, dispositions, and opportunities.

Applied Design Thinking is Strategic Innovation

Applied Design Thinking is Strategic Innovation

How we behave in order to apply design thinking successfully:

- We bid farewell to prejudices on “how things work.”
- We put aside expectations about what will happen.
- We strengthen our curiosity to understand facts and problems in depth.
- We open ourselves up to new possibilities.
- We ask simple questions.
- We try things out and learn from it.

A brief history and frameworks

A brief history and frameworks

- Design Thinking developed as an approach to problem-solving beginning in the 1950s within the industrial design, science, and technology communities. In the 1960s, inclusive and collaborative problem-solving began to replace a closed-off and selective approach. Gathering of user feedback in the design process became popular. Increased usage of computers and technology led to the start of human-computer interaction design in the 1970s.
- Beginning in the 1990s, a shift to specialization within the design process made its presence apparent as the so-called wicked problems in Design Thinking. In 1991, IDEO was formed as a company, resulting from a three-way merger, and invited diverse experts from anthropology, business strategy, and other disciplines to create design teams

A brief history and frameworks

- For the past 20 years, the Design Thinking approach and its ability to drive innovation in business and drive revenue has caught the eyes of many executives. A variety of frameworks have appeared that define the phases and steps used in Design Thinking engagements and workshops (as a search for “Design Thinking” via an Internet search engine will readily demonstrate).
- Each framework should be evaluated for its effectiveness in solving your problems and for your situations. Additionally, the methods and time spent within each phase of a framework might also be different from project to project. You could find that you might go in reverse or skip phases depending how the problem and information unfolds.

A brief history and frameworks

- Keep in mind that a framework is simply that. It should be flexible to bend to what you need and help to drive proper outcomes. Additionally, the methods within each phase of the framework should be just as flexible.
- Methods can be thought of as exercises or activities that produce insight or information. There are thousands of methods that can be applied within each phase, and each phase is subjective to the outcome you need to achieve. For example, in the Stanford d.school framework (described in the next subsection), you could use a persona, empathy map, interviews, surveys, and data on website visits or do shadowing to gain outcomes involving a group of people during the define phase.

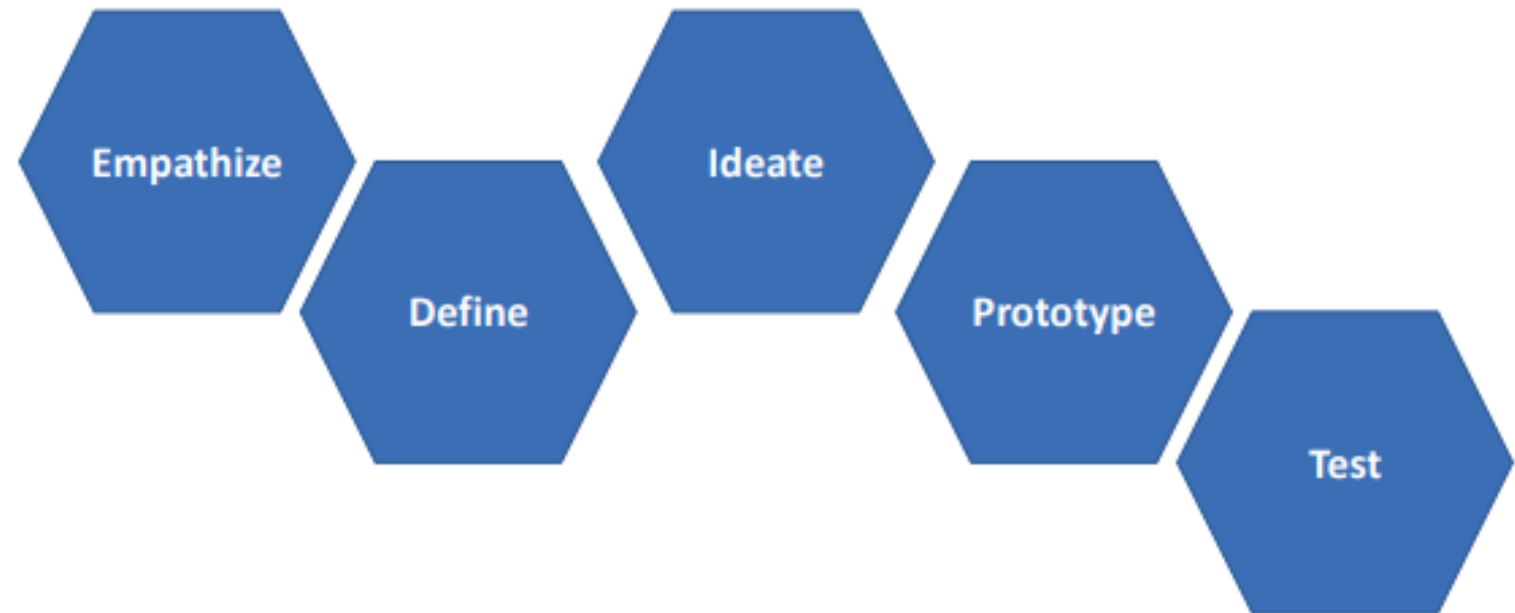
A brief history and frameworks

- Each method will yield only a portion of information needed. Interviews provide qualitative data such as sentiment or why someone does something.
- But you might not see hard numbers or facts. With surveys, you will get the numbers and quantitative facts, but not the why.
- Carefully consider what information you need before you invest time, money, and resource in a method that doesn't yield the full view of results and information you need for decision-making.

Stanford d.school Framework

- As you might expect given Stanford University's long history of teaching Design Thinking in its Institute of Design, it has defined one of the more popular frameworks being used. Key phases in the Stanford d.school framework are empathize, define, ideate, prototype, and test. These are illustrated in the diagram in Figure 1-2

Figure 1-2: Stanford d.school Design Thinking Framework



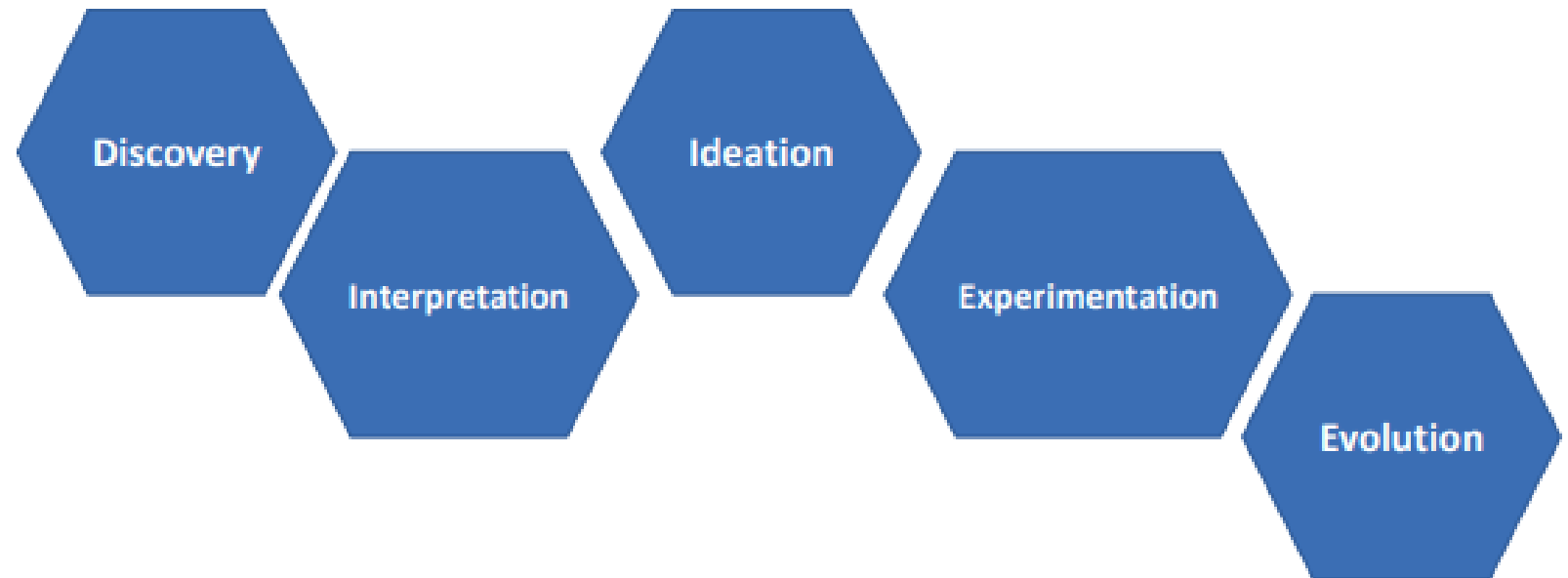
Stanford d.school Framework

- During the empathize phase, the focus is on interviews, shadowing, seeking to understand, and non-judgmental methods. In the define phase, the focus moves to understanding personas, role objectives, decisions, challenges, and pain points.
- In the ideate phase, ideas are shared, all ideas might be considered worthy, a diverge and converge method can be used, ideas might be extended using a "Yes and" method, and prioritization frequently takes place. During the prototype phase, mockups and storyboards might be created, a "keep it simple" approach might be applied, and methods of failing fast and quickly iterating might also be applied.
- In the testing phase, gaining an understanding of impediments, an understanding of what works, evaluating tests through role-plays, and performing tests in fast iterations are the methods that might be used.

IDEO Framework

- Another popular framework is IDEO's Design Thinking methodology. Key phases in this framework are discovery, interpretation, ideation, experimentation, and evolution. These are illustrated in Figure 1-3 using a representation like that in Figure 1-2 so that you can see the similarity to the Stanford d.school version

Figure 1-3: IDEO Design Thinking Framework



IDEO Framework

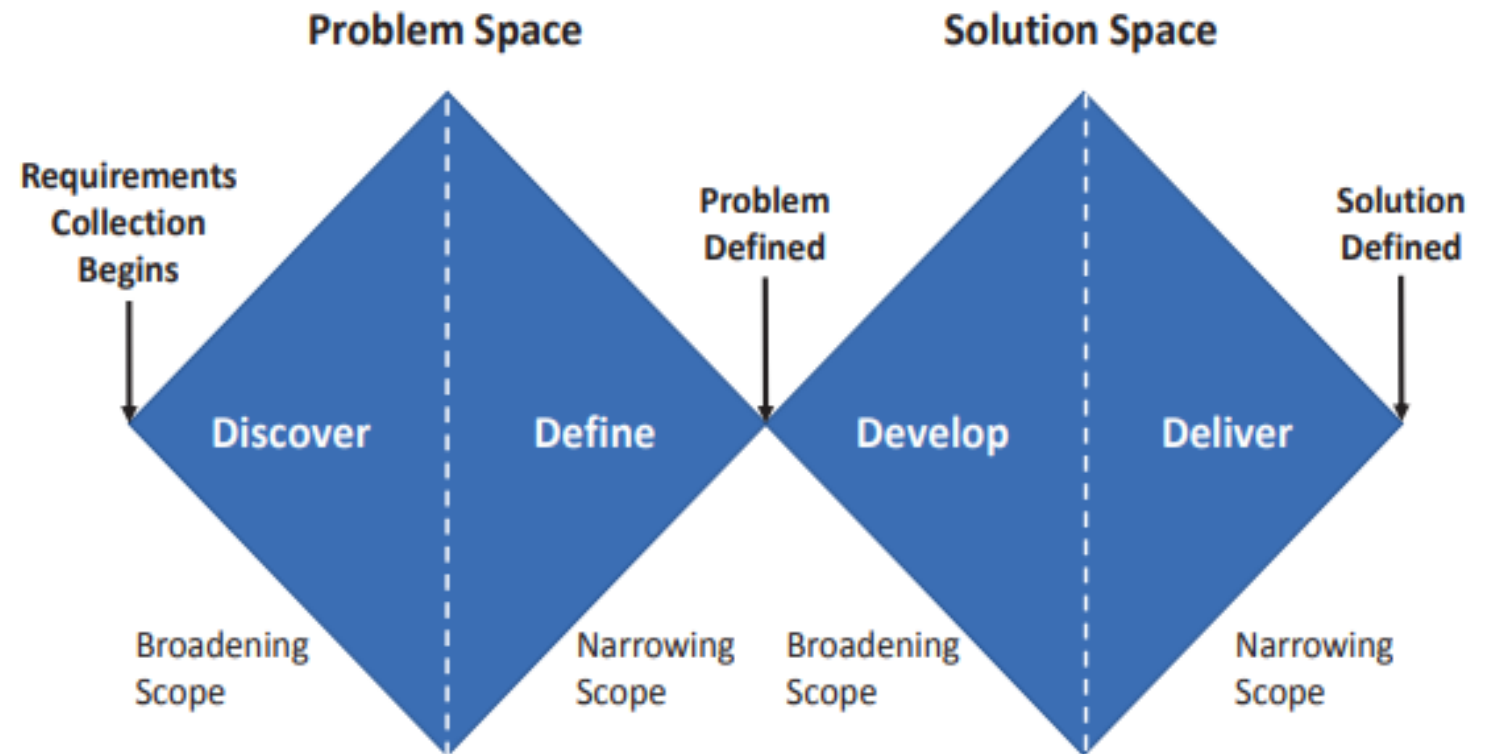
- During the discovery phase, the focus is on understanding the approach to a challenge by understanding what it is, preparing research, and gathering inspiration.
- In the interpretation phase, stories are told, there is a search for meaning, and opportunities are framed. In ideation, creation ideas are generated and refined.
- The experimentation phase consists of making prototypes and getting feedback.
- Evolution includes tracking learnings and moving forward

Double Diamond Design Methodology

Double Diamond Design Methodology

- At the heart of the original framework that most other innovative design frameworks are based upon is the UK Design Council's design methodology, the Double Diamond – a clear, comprehensive, and visual description of the design process. Launched in 2004, the Double Diamond has become world-renowned with millions of references to it on the Web.
- The Design Council's Double Diamond clearly conveys a design process to designers and non-designers alike. We illustrate the Double Diamond in Figure 1-4

Figure 1-4: Double
Diamond Design
Thinking
Methodology



Double Diamond Design Methodology

- The two diamonds represent the problem space and solution space in this approach. The left portion of each diamond represents a process of exploring an issue more widely or deeply (divergent thinking), while the right portion of each diamond represents taking focused action (convergent thinking). Key phases in the Double Diamond are discover, define, develop, and deliver.
- Discover and define are part of the problem space, while develop and deliver are part of the solution space. The phases are fixed within this methodology; however, the objectives or intent within these phases can change depending upon what we know (or don't know).

Double Diamond Design Methodology

- Within the problem space diamond, we gain an understanding, rather than simply assuming, what the problem is. We begin collecting requirements within the discover phase. Typical objectives in this phase include:
 - Setting scope – Defining an initial problem or vision statement
 - Determining people/stakeholders – Who is impacted or influenced
 - Determining what is the current state – Good or bad

Design Thinking, DevOps, and Adoption

Design Thinking, DevOps, and Adoption

- Design Thinking helps organizations discern unmet needs, create value from these insights, and create competitive advantages. Today, it is often used in conjunction with the DevOps approach adopted by many software and AI developers.
- While DevOps has helped teams of developers organize around their development challenges, Design Thinking helps businesses solve real business complex challenges using innovation. Change is hard for many, but Design Thinking gives order to chaos. It's a way to navigate, explore, and test ideas to ensuring value, business viability, and technical feasibility. Combining the two approaches fosters a collaborative and usercentered culture with an iterative approach to fail forward with change.

Design Thinking, DevOps, and Adoption

- In the past, innovation was often approached from the top down. This approach was omnidirectional and not very scalable at any level. Incorporating Design Thinking removes silos, invites healthy conflict in discussion to challenge the norms, and creates ideas that are valuable. These, in turn, help businesses create revenue and deliver customer and/or employee satisfaction.
- Many organizations do not do this well. The transition to Design Thinking needs to be driven from an executive level to help build innovative design into an organization's DNA. It should trickle down into every software or AI project being considered along with the challenges those project charters are trying to solve.

Design Thinking, DevOps, and Adoption

- How long might such cultural change take? Leveraging some earlier work by Jakob Nielsen, Gena Drahun estimated the average time it would take to develop a user-driven culture in 2015. According to Drahun, the typical phases and adoption periods are as follows:
 - Stage 1 – Developer-centered
 - Stage 2 – Skunkworks (2 years)
 - Stage 3 – Dedicated budget (4 years)
 - Stage 4 – Managed (7 years)
 - Stage 5 – Systematic process (13 years)
 - Stage 6 – Integrated user-centered design (20 years)
 - Stage 7 – User-driven corporation (40 years)

Design Thinking, DevOps, and Adoption

- However, in the authors' view, proven benefits from leveraging a Design Thinking approach (and the necessity to react more quickly to emerging business challenges) are helping to speed adoption in many organizations. In Chapter 7, we describe how change management is an important part of speeding this adoption.
- One advantage to using a Design Thinking approach is that it helps an organization more deeply understand the root cause of problems, the contributing factors, context, and reasons for the problem. Learning about the cause can sometimes point to a solution. Such insight may not have been available during earlier approaches used in addressing the problem.

Design Thinking, DevOps, and Adoption

- Design Thinking also creates a sense of ownership for the teams working in collaboration to help solve the problem. This unified approach can create alignment. Specifically, as the team works together to understand and clarify various points of information, perspectives, opinions, and thoughts, they can synthesize the information in to a collective and more holistic point of view.

Design Thinking, DevOps, and Adoption

- Additionally, Design Thinking ensures that the final product, service, or solution meets the initial objectives or client requirements. Since part of the process is defining success and testing if ideas deliver success, the value of results is ensured. Since Design Thinking is iterative, continuous iterative loops through changing information, ideation, validation, and implementation can result in a continuous improvement process that builds on the success and failure of the last iteration.
- The ultimate benefit is the solution continually gets better as more information, knowledge, and ideas are applied.

Knowledge Check

Who is Design thinkers?

- a. People who can solve all problems.
- b. People who can solve complex problems.
- c. People who can solve specific problems.
- d. People who can create anything.

Knowledge Check

IDEO Design Thinking Framework, except:

- a. Discover
- b. Ideation
- c. Evolution
- d. Implementation

Knowledge Check

How long the typical phases of Design Thinking, DevOps, and Adoption for stage 5: Systematic process?

- a. 7 Years
- b. 13 Years
- c. 20 Years
- d. 40 Years

Activities

- Make a group of 3-4 person/group.
- Define a problem and solution using Double Diamond Design Methodology.
- Discuss it with your group and present it.

Summary

- As we come to the end of this chapter, you should better understand how Design Thinking helps organizations identify and solve problems more rapidly and helps drive innovation. You should also see how this iterative approach provides a means to get beyond setbacks that occur in projects and use those setbacks as learning experiences.
- You should now recognize some keys to taking a productive approach (as well as some of the non-productive methods and approaches that should be avoided). You also had a brief introduction to popular frameworks that help ensure a productive approach will be taken.

Summary

- Finally, you should now think of Design Thinking as complementary to a modern DevOps approach. And you should understand its benefits and how those can help drive adoption. We begin a deeper exploration of applying the methodology in



Thank You